

Lemma: 对 $\forall z = x + yi \in \mathbb{C}$ (其中 $x, y \in \mathbb{R}$), 有:

$$z \neq 0 \iff x^2 + y^2 > 0 \iff \sqrt{x^2 + y^2} > 0$$

Proof: 若 $z \neq 0$, 则 x 与 y 不同时为 0 $\therefore x^2 + y^2 > 0$.

$$\text{若 } x^2 + y^2 > 0, \text{ 则 } (x-0)^2 + (y-0)^2 > 0 \therefore \sqrt{(x-0)^2 + (y-0)^2} > 0$$

$$\therefore z \neq 0.$$

(或: 假设 $z=0$. 则 $x=y=0$. 与 $x^2+y^2>0$ 矛盾.) $\square$

Lemma (复数的指数形式). 对 $\forall z \in \mathbb{C} \setminus \{0\}$, $\exists r \in \mathbb{R}_{>0}$ ~~和~~ $\theta \in \mathbb{R}$, 使得 $z = re^{i\theta}$.

Proof: $\because z \in \mathbb{C} \setminus \{0\}$. $\therefore z = x + yi$, 其中 x 和 y 不同时为 0, $x, y \in \mathbb{R}$.

$$\therefore x^2 + y^2 > 0 \quad \therefore \sqrt{x^2 + y^2} > 0.$$

$$\begin{aligned} \therefore z = x + yi &= (\sqrt{x^2 + y^2} + 0i) \left(\frac{x}{\sqrt{x^2 + y^2}} + \frac{y}{\sqrt{x^2 + y^2}} i \right) \\ &= \sqrt{x^2 + y^2} \cdot \left(\frac{x}{\sqrt{x^2 + y^2}} + \frac{y}{\sqrt{x^2 + y^2}} i \right) \end{aligned}$$

↓
复数乘法

$$\text{令 } r = \sqrt{x^2 + y^2} \quad \therefore r \in \mathbb{R}_{>0}. \quad \exists \theta \in \mathbb{R}, \text{ s.t. } \cos \theta = \frac{x}{\sqrt{x^2 + y^2}}, \sin \theta = \frac{y}{\sqrt{x^2 + y^2}}$$

$$\therefore z = r \cdot (\cos \theta + i \sin \theta) = re^{i\theta}$$

↓
复数乘法

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直接用欧拉公式.

$$\therefore z = re^{i\theta}, \quad r \in \mathbb{R}_{>0} \text{ 且 } \theta \in \mathbb{R}. \quad \square$$

Lemma: 对 $\forall z_1 = r_1 e^{i\theta_1} \in \mathbb{C}$, $z_2 = r_2 e^{i\theta_2} \in \mathbb{C}$ (其中 $r_1, r_2 \in \mathbb{R}_{>0}$, $\theta_1, \theta_2 \in \mathbb{R}$)

有: $z_1 \cdot z_2 = r_1 r_2 e^{i(\theta_1 + \theta_2)}$

Proof: 当 z_1 或 z_2 为 0 时, 取 r_1 或 r_2 为 0. θ_1 和 θ_2 是任意实数

$$\begin{aligned} z_1 \cdot z_2 &= r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 (\cos\theta_1 + i\sin\theta_1) \cdot r_2 (\cos\theta_2 + i\sin\theta_2) \\ &= (r_1 \cos\theta_1 + (r_1 \sin\theta_1) i) \cdot ((r_2 \cos\theta_2) + (r_2 \sin\theta_2) i) \\ &= (r_1 \cos\theta_1 r_2 \cos\theta_2 - r_1 \sin\theta_1 r_2 \sin\theta_2) + (r_1 \cos\theta_1 r_2 \sin\theta_2 + r_2 \cos\theta_2 r_1 \sin\theta_1) i \\ &= (r_1 r_2 \cos(\theta_1 + \theta_2)) + (r_1 r_2 \sin(\theta_1 + \theta_2)) i \\ &= (r_1 r_2 + 0i) (\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)) \\ &= r_1 r_2 e^{i(\theta_1 + \theta_2)} \end{aligned}$$

定理 (De Moivre's formula) 对 $\forall z = re^{i\theta} \in \mathbb{C}$, $\forall n \in \mathbb{Z}$, 有:

$$z^n = (re^{i\theta})^n = r^n e^{i(n\theta)}$$

Proof: $\because z = re^{i\theta} \therefore z^n = (re^{i\theta})^n$. 左边的等号显然.

当 $n=0$ 时, 要求 $z \neq 0$. $z^0 = 1$. $\because z \neq 0 \therefore r \in \mathbb{R}_{>0}$.

$$\therefore r^0 e^{i(0 \cdot \theta)} = 1 \cdot e^{i0} = 1 \cdot (\cos 0 + i\sin 0) = 1 \cdot (1 + 0i) = 1$$

$$\text{当 } n=1 \text{ 时, } (re^{i\theta})^1 = re^{i\theta} = r^1 e^{i(1 \cdot \theta)}$$

$$\text{当 } n=2 \text{ 时, } (re^{i\theta})^2 = (re^{i\theta})(re^{i\theta}) = r^2 e^{i(2\theta)}$$

假设当 $n=k \in \mathbb{Z}_{\geq 2}$ 时, 有: $(re^{i\theta})^k = r^k e^{i(k\theta)}$. 则当 $n=k+1$ 时,

$$(re^{i\theta})^{k+1} = (re^{i\theta})^k (re^{i\theta}) = (r^k e^{i(k\theta)}) (re^{i\theta}) = r^{k+1} e^{i((k+1)\theta)}$$

$$\therefore \text{对 } \forall n \in \mathbb{Z}_{\geq 0}, \text{ 有: } (re^{i\theta})^n = r^n e^{i(n\theta)}$$

$$\text{对 } \forall n \in \mathbb{Z}_{\leq -1}, \text{ 有: } -n \in \mathbb{Z}_{\geq 1} \quad \therefore (re^{i\theta})^n = (re^{i\theta})^{-(-n)} = \frac{1}{(re^{i\theta})^{-n}}$$

$$= \frac{1}{r^{-n} e^{i(-n)\theta}} = \frac{1}{r^{-n} (\cos((-n)\theta) + i\sin((-n)\theta))}$$

$$= \frac{1}{r^{-n}} \cdot \frac{\cos((-n)\theta) - i\sin((-n)\theta)}{1} = r^{-(-n)} (\cos(n\theta) + i\sin(n\theta))$$

$$= r^n (\cos(n\theta) + i\sin(n\theta)) = r^n e^{i(n\theta)} \quad \square$$

Lemma: $n \in \mathbb{Z}_{\geq 1}$. 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d|n$, 定义两个集合:

$$J_d = \{r \in \{1, 2, \dots, n\} \mid \frac{n}{\gcd(r, n)} = d\}$$

$$K_d = \{k \in \mathbb{Z}_{\geq 1} \mid k \leq d \text{ 且 } \gcd(k, d) = 1\}$$

$$\text{则有: } |J_d| = |K_d| = \varphi(d)$$

Proof: 定义映射 $f: K_d \rightarrow J_d$

$$k \mapsto k \cdot \frac{n}{d}$$

$$\therefore d|n \quad \therefore \frac{n}{d} \in \mathbb{Z} \quad \therefore n \in \mathbb{Z}_{\geq 1} \text{ 且 } d \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{n}{d} \in \mathbb{Z}_{\geq 1}$$

$$\text{对 } \forall k \in K_d, \text{ 有: } k \in \mathbb{Z}_{\geq 1} \text{ 且 } k \leq d \text{ 且 } \gcd(k, d) = 1$$

$$\therefore k \in \mathbb{Z}_{\geq 1} \text{ 且 } \frac{n}{d} \in \mathbb{Z}_{\geq 1} \quad \therefore k \cdot \frac{n}{d} \in \mathbb{Z}_{\geq 1}$$

$$\therefore k \leq d \text{ 且 } \frac{n}{d} > 0 \quad \therefore k \cdot \frac{n}{d} \leq d \cdot \frac{n}{d} = n$$

$$\therefore f(k) = k \cdot \frac{n}{d} \in \{1, \dots, n\}$$

$$\therefore \gcd(f(k), n) = \gcd\left(k \cdot \frac{n}{d}, d \cdot \frac{n}{d}\right) = \frac{n}{d} \gcd(k, d) = \frac{n}{d}$$

$$\therefore \frac{n}{\gcd(f(k), n)} = \frac{n}{\frac{n}{d}} = d. \quad \therefore f(k) \in \{1, \dots, n\} \text{ 且 } \frac{n}{\gcd(f(k), n)} = d$$

$$\therefore f(k) \in J_d \quad \therefore f(K_d) \subseteq J_d.$$

$$\text{对 } \forall k_1, k_2 \in K_d, \text{ 有: } k_1, k_2 \in \mathbb{Z}_{\geq 1}, k_1 \leq d, k_2 \leq d, \\ \gcd(k_1, d) = 1, \gcd(k_2, d) = 1$$

$$\text{若 } k_1 = k_2, \text{ 则 } \therefore k_1 = k_2, k_1, k_2 \in \mathbb{Z}_{\geq 1}, \frac{n}{d} \in \mathbb{Z}_{\geq 1}$$

$$\therefore f(k_1) = k_1 \cdot \frac{n}{d} = k_2 \cdot \frac{n}{d} = f(k_2)$$

$$\therefore f: K_d \rightarrow J_d \text{ 是一个映射.}$$

$$\text{若 } f(k_1) = f(k_2), \text{ 则 } k_1 \cdot \frac{n}{d} = k_2 \cdot \frac{n}{d}. \quad \therefore \frac{n}{d} \in \mathbb{Z}_{\geq 1}$$

$$\therefore k_1 = k_2 \quad \therefore f: K_d \rightarrow J_d \text{ 是一个单射.}$$

$$\text{对 } \forall r \in J_d, \text{ 有: } r \in \{1, \dots, n\} \text{ 且 } \frac{n}{\gcd(r, n)} = d.$$

$$\therefore r \in \mathbb{Z}_{\geq 1} \text{ 且 } n \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd(r, n) \in \mathbb{Z}_{\geq 1}. \quad \therefore d \in \mathbb{Z}_{\geq 1}$$

$$\therefore \frac{n}{d} = \gcd(r, n) \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd(r, n) \mid r \quad \therefore \frac{n}{d} \mid r.$$

$$\therefore r \in \mathbb{Z}_{\geq 1} \text{ 且 } \frac{n}{d} \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{r}{\frac{n}{d}} \in \mathbb{Z}_{\geq 1} \quad \text{令 } k = \frac{r}{\frac{n}{d}}$$

$$\therefore k \in \mathbb{Z}_{\geq 1} \quad \therefore k = \frac{r}{\frac{n}{d}}, \frac{n}{d} \in \mathbb{Z}_{\geq 1} \quad \therefore k \cdot \frac{n}{d} = r \leq n$$

$$\therefore k \cdot \frac{n}{d} \leq n. \quad \therefore n \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{k}{d} \leq 1 \quad \therefore d \in \mathbb{Z}_{\geq 1} \quad \therefore k \leq d.$$

$$\because \gcd(r, n) = \frac{n}{d} \quad \therefore \frac{n}{d} = \gcd(r, n) = \gcd(k \cdot \frac{n}{d}, d \cdot \frac{n}{d})$$

$$\circledast \because \frac{n}{d} \in \mathbb{Z}_{\geq 1}, \quad k \in \mathbb{Z}_{\geq 1}, \quad d \in \mathbb{Z}_{\geq 1}$$

$$\therefore \frac{n}{d} = \gcd(k \cdot \frac{n}{d}, d \cdot \frac{n}{d}) = \frac{n}{d} \gcd(k, d) \quad \therefore \gcd(k, d) = 1$$

$$\therefore k \in \mathbb{Z}_{\geq 1} \text{ 且 } k \leq d \text{ 且 } \gcd(k, d) = 1 \quad \therefore k \in K_d.$$

$$\# \because f(k) = k \cdot \frac{n}{d} = r \quad \therefore f: K_d \rightarrow J_d \text{ 是一个满射.}$$

$$\therefore f: K_d \rightarrow J_d \text{ 是一个双射.}$$

$$\therefore |K_d| = |J_d| \quad \therefore |J_d| = |K_d| = \varphi(d) \quad \square$$

$$\text{Remark: } \because 1 \in \mathbb{Z}_{\geq 1}, \quad 1 \leq d, \quad \gcd(1, d) = 1$$

$$\therefore 1 \in K_d \quad \therefore K_d \neq \emptyset.$$

$$\because d \in \mathbb{Z}_{\geq 1} \quad \therefore 0 < \frac{1}{d} \leq 1 \quad \because n \in \mathbb{Z}_{\geq 1} \quad \therefore n > 0$$

$$\therefore \frac{n}{d} \leq n. \quad \because d | n \quad \therefore \frac{n}{d} \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{n}{d} \in \{1, 2, \dots, n\}$$

$$\therefore \gcd\left(\frac{n}{d}, n\right) = \gcd\left(\frac{n}{d} \cdot 1, \frac{n}{d} \cdot d\right) = \frac{n}{d} \gcd(1, d) = \frac{n}{d}$$

$$\therefore \frac{n}{\gcd(\frac{n}{d}, n)} = \frac{n}{\frac{n}{d}} = n \cdot \frac{d}{n} = d$$

$$\therefore \frac{n}{d} \in \{1, 2, \dots, n\} \text{ 且 } \frac{n}{\gcd(\frac{n}{d}, n)} = d \quad \therefore \frac{n}{d} \in J_d \quad \therefore J_d \neq \emptyset$$

Lemma: $n \in \mathbb{Z}_{\geq 1}$, 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d | n$, 定义集合

$$J_d = \{r \in \{1, 2, \dots, n\} \mid \frac{n}{\gcd(r, n)} = d\}. \quad \text{则有:}$$

$$\textcircled{1}. \text{ 对 } \forall d \in \mathbb{Z}_{\geq 1} \text{ 且 } d | n, \text{ 有: } \frac{n}{d} \in J_d. \text{ 因此 } J_d \neq \emptyset. \quad J_d \subseteq \{1, 2, \dots, n\}.$$

$$\textcircled{2} \{1, 2, \dots, n\} = \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d$$

Proof: 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d|n$, 有: $\because d|n, d \in \mathbb{Z}_{\geq 1}, n \in \mathbb{Z}_{\geq 1}$

$$\therefore \frac{n}{d} \in \mathbb{Z}_{\geq 1} \quad \because d \in \mathbb{Z}_{\geq 1} \quad \therefore 0 < \frac{1}{d} \leq 1 \quad \because n > 0$$

$$\therefore \frac{n}{d} \leq n \quad \therefore \frac{n}{d} \in \{1, 2, \dots, n\}$$

$$\therefore \gcd\left(\frac{n}{d}, n\right) = \gcd\left(\frac{n}{d} \cdot 1, \frac{n}{d} \cdot d\right) = \frac{n}{d} \gcd(1, d) = \frac{n}{d}$$

$$\therefore \frac{n}{\gcd(\frac{n}{d}, n)} = \frac{n}{\frac{n}{d}} = n \cdot \frac{d}{n} = d$$

$$\therefore \frac{n}{d} \in \{1, 2, \dots, n\} \text{ 且 } \frac{n}{\gcd(\frac{n}{d}, n)} = d \quad \therefore \frac{n}{d} \in J_d \quad \therefore J_d \neq \emptyset$$

显然 $J_d \subseteq \{1, 2, \dots, n\}$. ①得证.

$\therefore$ 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d|n$, 有: $J_d \subseteq \{1, 2, \dots, n\}$.

$$\therefore \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d \subseteq \{1, 2, \dots, n\}$$

对 $\forall r \in \{1, 2, \dots, n\}$, 有: $\because r \in \mathbb{Z}_{\geq 1}$ 且 $n \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd(r, n) \in \mathbb{Z}_{\geq 1}$
 $\because \gcd(r, n) | n \quad \therefore \frac{n}{\gcd(r, n)} \in \mathbb{Z}_{\geq 1} \quad \text{令 } d = \frac{n}{\gcd(r, n)} \quad \therefore d \in \mathbb{Z}_{\geq 1}$

$$\therefore d = \frac{n}{\gcd(r, n)}, d \in \mathbb{Z}_{\geq 1}, \gcd(r, n) \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{n}{d} = \gcd(r, n) \in \mathbb{Z}_{\geq 1}$$

$$\therefore d | n. \quad \therefore d | n \text{ 且 } d \in \mathbb{Z}_{\geq 1}. \quad \therefore r \in \{1, 2, \dots, n\} \text{ 且 } \frac{n}{\gcd(r, n)} = d$$

$$\therefore r \in J_d \quad \therefore r \in \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d \quad \therefore \{1, 2, \dots, n\} \subseteq \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d$$

$$\therefore \{1, 2, \dots, n\} = \bigcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d$$

又对 $\forall d_1, d_2 \in \mathbb{Z}_{\geq 1}$, $d_1 | n$, $d_2 | n$, $d_1 \neq d_2$,

假设 $J_{d_1} \cap J_{d_2} \neq \emptyset$. 则 $\exists r \in J_{d_1} \cap J_{d_2} \therefore r \in J_{d_1} \text{ 且 } r \in J_{d_2}$

$$\therefore r \in J_{d_1} \quad \therefore r \in \{1, 2, \dots, n\} \text{ 且 } \frac{n}{\gcd(r, n)} = d_1$$

$$\therefore r \in J_{d_2} \quad \therefore r \in \{1, 2, \dots, n\} \text{ 且 } \frac{n}{\gcd(r, n)} = d_2 \quad \therefore d_1 = d_2. \text{ 矛盾.}$$

$$\therefore J_{d_1} \cap J_{d_2} = \emptyset. \quad \therefore \{1, 2, \dots, n\} = \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d \quad \square$$

推论: $\sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) = n$. (其中 $n \in \mathbb{Z}_{\geq 1}$).

Proof: $\therefore n \in \mathbb{Z}_{\geq 1} \quad \therefore$ 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d | n$, 定义集合

$$J_d = \left\{ r \in \{1, 2, \dots, n\} \mid \frac{n}{\gcd(r, n)} = d \right\}. \quad \therefore \{1, 2, \dots, n\} = \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d$$

$$\therefore n = |\{1, 2, \dots, n\}| = \left| \bigsqcup_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} J_d \right| = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} |J_d| = \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d)$$

$$\therefore \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1}}} \varphi(d) = n. \quad \square$$

规定: $\sum_{k \in \emptyset} a_k = 0$ ($a_k \in \mathbb{C}$), $\prod_{k \in \emptyset} a_k = 1$ ($a_k \in \mathbb{C}$)

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, $\prod_{k=1}^n (1+X^k) \in \mathbb{C}[X]$. 则有:

$$\prod_{k=1}^n (1+X^k) = \sum_{S \subseteq \{1, \dots, n\}} X^{\sum_{k \in S} k}$$

Proof: 当 $n=1$ 时, 左边 = $\prod_{k=1}^1 (1+X^k) = 1+X^1 = 1+X$

$$\text{右边} = \sum_{S \subseteq \{1\}} X^{\sum_{k \in S} k} = X^{\sum_{k \in \emptyset} k} + X^{\sum_{k \in \{1\}} k} = X^0 + X^1 = 1+X$$

$\therefore$ 左边 = 右边.

当 $n=2$ 时, 左边 = $\prod_{k=1}^2 (1+X^k) = (1+X^1)(1+X^2) = (1+X)(1+X^2)$

$$= 1+X+X^2+X^3$$

$$\text{右边} = \sum_{S \subseteq \{1, 2\}} X^{\sum_{k \in S} k} = X^{\sum_{k \in \emptyset} k} + X^{\sum_{k \in \{1\}} k} + X^{\sum_{k \in \{2\}} k} + X^{\sum_{k \in \{1, 2\}} k}$$

$$= X^0 + X^1 + X^2 + X^3 = 1+X+X^2+X^3 \quad \therefore \text{左边} = \text{右边}.$$

当 $n=3$ 时, 左边 = $\prod_{k=1}^3 (1+X^k) = (1+X^1)(1+X^2)(1+X^3)$

$$= (1+X+X^2+X^3)(1+X^3) = 1+X+X^2+2X^3+X^4+X^5+X^6$$

$$\text{右边} = \sum_{S \subseteq \{1, 2, 3\}} X^{\sum_{k \in S} k} = X^{\sum_{k \in \emptyset} k} + X^{\sum_{k \in \{1\}} k} + X^{\sum_{k \in \{2\}} k} + X^{\sum_{k \in \{3\}} k} + X^{\sum_{k \in \{1, 2\}} k} + X^{\sum_{k \in \{1, 3\}} k} + X^{\sum_{k \in \{2, 3\}} k} + X^{\sum_{k \in \{1, 2, 3\}} k}$$

$$+ X^{\sum_{k \in \{1,3\}} k} + X^{\sum_{k \in \{2,3\}} k} + X^{\sum_{k \in \{1,2,3\}} k} = X^0 + X^1 + X^2 + X^3 + X^3 + X^4 + X^5 + X^6 = 1 + X + X^2 + 2X^3 + X^4 + X^5 + X^6 \quad \therefore \text{左边} = \text{右边}.$$

假设对于 $n = m \in \mathbb{Z}_{\geq 3}$ 时, 结论成立, 即:

$$\prod_{k=1}^m (1 + X^k) = \sum_{S \subseteq \{1, \dots, m\}} X^{\sum_{k \in S} k}$$

$$\text{则当 } n = m+1 \text{ 时, 左边} = \prod_{k=1}^{m+1} (1 + X^k) = \left(\prod_{k=1}^m (1 + X^k) \right) \cdot (1 + X^{m+1})$$

$$= \left(\sum_{S \subseteq \{1, \dots, m\}} X^{\sum_{k \in S} k} \right) \cdot (1 + X^{m+1})$$

$$= \sum_{S \subseteq \{1, \dots, m\}} X^{\sum_{k \in S} k} + \sum_{S \subseteq \{1, \dots, m\}} X^{\sum_{k \in S} k + (m+1)}$$

$$= \sum_{S \subseteq \{1, \dots, m\}} X^{\sum_{k \in S} k} + \sum_{S \subseteq \{1, \dots, m\}} X^{\sum_{k \in S \cup \{m+1\}} k}$$

$$= \sum_{\substack{S \subseteq \{1, \dots, m, m+1\} \\ m+1 \notin S}} X^{\sum_{k \in S} k} + \sum_{\substack{S \subseteq \{1, \dots, m, m+1\} \\ m+1 \in S}} X^{\sum_{k \in S} k} = \sum_{S \subseteq \{1, \dots, m, m+1\}} X^{\sum_{k \in S} k} = \text{右边}$$

$$\therefore \text{对 } \forall n \in \mathbb{Z}_{\geq 1}, \text{ 有: } \prod_{k=1}^n (1 + X^k) = \sum_{S \subseteq \{1, \dots, n\}} X^{\sum_{k \in S} k} \quad \square$$

Remark: 这道题是数学归纳法很好的入门练习.

Lemma: X 和 Y 是任意的两个集合, $P(X) = \{D \mid D \subseteq X\}$,
 $P(Y) = \{D \mid D \subseteq Y\}$. $f: X \rightarrow Y$ 是任意的一个从 X 到 Y 的映射.

定义映射 $F: P(X) \rightarrow P(Y)$

$$A \mapsto \{f(x) \mid x \in A\}.$$

则有: ① 映射 $F: P(X) \rightarrow P(Y)$ 良定义

② $f: X \rightarrow Y$ 是单射 $\Leftrightarrow F: P(X) \rightarrow P(Y)$ 是单射

③ $f: X \rightarrow Y$ 是满射 $\Leftrightarrow F: P(X) \rightarrow P(Y)$ 是满射

④ $f: X \rightarrow Y$ 是双射 $\Leftrightarrow F: P(X) \rightarrow P(Y)$ 是双射.

Proof: 对 $\forall A \in P(X)$, 有: $A \subseteq X$. 对 $\forall x \in A$, 有: $x \in A \subseteq X$

$$\therefore f(x) \in Y \quad \therefore \{f(x) \mid x \in A\} \subseteq Y \quad \therefore \{f(x) \mid x \in A\} \in P(Y)$$

$$\therefore F(A) = \{f(x) \mid x \in A\} \in P(Y)$$

对 $\forall A_1, A_2 \in P(X)$, 有: $A_1 \subseteq X$ 且 $A_2 \subseteq X$.

$$\text{若 } A_1 = A_2, \text{ 则 } \because A_1 = A_2 \subseteq X \quad \therefore \{f(x) \mid x \in A_1\} = \{f(x) \mid x \in A_2\}.$$

$$\therefore F(A_1) = \{f(x) \mid x \in A_1\} = \{f(x) \mid x \in A_2\} = F(A_2)$$

$\therefore F: P(X) \rightarrow P(Y)$ 是一个良定义的映射. ① 得证.

② $\Rightarrow$): 对 $\forall A_1, A_2 \in P(X)$, 有: $A_1 \subseteq X$ 且 $A_2 \subseteq X$.

$$\text{若 } F(A_1) = F(A_2), \text{ 则 } \{f(x) \mid x \in A_1\} = \{f(x) \mid x \in A_2\}.$$

对 $\forall \alpha \in A_1$, 有: $f(\alpha) \in \{f(x) | x \in A_1\} = \{f(x) | x \in A_2\}$

$\therefore \exists \beta \in A_2$, s.t. $f(\alpha) = f(\beta) \quad \because f \text{ 是单射} \quad \therefore \alpha = \beta \in A_2$

$\therefore A_1 \subseteq A_2$. 同理可证 $A_2 \subseteq A_1 \quad \therefore A_1 = A_2$

$\therefore F: P(X) \rightarrow P(Y)$ 是单射.

~~② ($\Leftarrow$): 对 $\forall \alpha, \beta \in X$, 有: $\{\alpha\}, \{\beta\} \subseteq X$.~~

~~$\therefore \{\alpha\}, \{\beta\} \in P(X) \quad \therefore F: P(X) \rightarrow P(Y)$ 是~~

② ($\Leftarrow$): 对 $\forall \alpha, \beta \in X$, 若 $f(\alpha) = f(\beta)$, 则有:

$\therefore \{\alpha\}, \{\beta\} \subseteq X \quad \therefore \{\alpha\}, \{\beta\} \in P(X)$.

~~$\therefore F: P(X) \rightarrow P(Y)$ 是单射~~

$\therefore F(\{\alpha\}) = \{f(x) | x \in \{\alpha\}\} = \{f(\alpha)\}$

$F(\{\beta\}) = \{f(x) | x \in \{\beta\}\} = \{f(\beta)\}$

$\because f(\alpha) = f(\beta) \quad \therefore F(\{\alpha\}) = F(\{\beta\})$

$\therefore F: P(X) \rightarrow P(Y)$ 是单射 $\therefore \{\alpha\} = \{\beta\} \quad \therefore \alpha = \beta$

$\therefore f: X \rightarrow Y$ 是单射.

③ ($\Rightarrow$): 对 $\forall B \in P(Y)$, 有: $B \subseteq Y$. $\because f: X \rightarrow Y$ 是映射.

$\therefore f^{-1}(B) \subseteq X \quad \therefore f^{-1}(B) = \{x \in X | f(x) \in B\} \subseteq X$.

$\therefore f^{-1}(B) \in P(X) \quad \therefore F(f^{-1}(B)) = \{f(x) | x \in f^{-1}(B)\}$

任取 $\{f(x) | x \in f^{-1}(B)\}$ 中的一元: $f(x) (x \in f^{-1}(B)) \quad \therefore x \in f^{-1}(B)$

$$\therefore x \in X \text{ 且 } f(x) \in B. \quad \therefore \{f(x) \mid x \in f^{-1}(B)\} \subseteq B.$$

$$\text{对 } \forall \beta \in B, \quad \therefore \beta \in B \subseteq Y \quad \therefore \beta \in Y. \quad \therefore f: X \rightarrow Y \text{ 是满射.}$$

$$\therefore \exists \alpha \in X, \text{ s.t. } f(\alpha) = \beta \in B. \quad \therefore \alpha \in f^{-1}(B)$$

$$\therefore f(\alpha) \in \{f(x) \mid x \in f^{-1}(B)\}. \quad \therefore \beta \in \{f(x) \mid x \in f^{-1}(B)\}.$$

$$\therefore B \subseteq \{f(x) \mid x \in f^{-1}(B)\} \quad \therefore \{f(x) \mid x \in f^{-1}(B)\} = B$$

$$\therefore F(f^{-1}(B)) = B \quad \therefore F: P(X) \rightarrow P(Y) \text{ 是满射.}$$

$$\textcircled{3}(\Leftarrow): \text{对 } \forall \beta \in Y, \text{ 有: } \{\beta\} \subseteq Y \quad \therefore \{\beta\} \in P(Y).$$

$$\therefore F: P(X) \rightarrow P(Y) \text{ 是满射, } \{\beta\} \in P(Y).$$

$$\therefore \exists A \in P(X), \text{ s.t. } F(A) = \{\beta\}.$$

$$\therefore \{\beta\} = F(A) = \{f(x) \mid x \in A\}.$$

$$\text{假设 } A = \emptyset, \text{ 则 } F(A) = \emptyset. \text{ 矛盾. } \therefore A \neq \emptyset.$$

$$\therefore \exists \alpha \in A. \quad \therefore \alpha \in A \text{ 且 } f(\alpha) \in \{f(x) \mid x \in A\} = \{\beta\}.$$

$$\therefore \alpha \in A \text{ 且 } f(\alpha) = \beta. \quad \therefore \alpha \in A \subseteq X \text{ 且 } f(\alpha) = \beta.$$

$$\therefore \alpha \in X \text{ 且 } f(\alpha) = \beta. \quad \therefore f: X \rightarrow Y \text{ 是满射.}$$

$$\textcircled{4} \text{ 由 } \textcircled{2}, \textcircled{1} \text{ 显然. } \square.$$

Lemma: X 和 Y 是任意的两个集合, $f: X \rightarrow Y$ 是一个单射,
 $A \subseteq X$, $B \subseteq Y$. 定义映射 $g: A \rightarrow B$ 则有:
 $x \mapsto f(x)$.

映射 $g: A \rightarrow B$ 良定且是双射 $\Leftrightarrow f(A) = \{f(x) | x \in A\} = B$.

Proof: $(\Rightarrow)$: 对 $\forall \beta \in \{f(x) | x \in A\}$, 有: $\exists \alpha \in A$, s.t. $\beta = f(\alpha)$.

$\because \alpha \in A$ 且映射 $g: A \rightarrow B$ 良定 $\therefore g(\alpha) \in B$

$\therefore f(\alpha) \in B \quad \because \beta \in B \quad \therefore \{f(x) | x \in A\} \subseteq B$

对 $\forall \beta \in B$, $\because g: A \rightarrow B$ 是双射 $\therefore g: A \rightarrow B$ 是满射

$\therefore \exists \alpha \in A$, s.t. $g(\alpha) = \beta \quad \therefore f(\alpha) = \beta$.

$\therefore \beta = f(\alpha)$ 且 $\alpha \in A \quad \therefore \beta \in \{f(x) | x \in A\}$.

$\therefore B \subseteq \{f(x) | x \in A\} \quad \therefore f(A) = \{f(x) | x \in A\} = B$.

$(\Leftarrow)$: 对 $\forall x \in A$, 有: $\because A \subseteq X \quad \therefore x \in X. \quad \therefore f(x) \in Y$.

$\because x \in A \quad \therefore f(x) \in f(A) = B \quad \therefore f(x) \in B$.

$\therefore g(x) = f(x) \in B \quad \therefore g(x) \in B. \quad \therefore g(A) \subseteq B$.

对 $\forall x_1, x_2 \in A$, 若 $x_1 = x_2$, 则 $\because A \subseteq X \quad \therefore x_1, x_2 \in X$.

$\because f: X \rightarrow Y$ 是映射, $x_1 = x_2 \in X \quad \therefore f(x_1) = f(x_2)$

$\therefore g(x_1) = f(x_1) = f(x_2) = g(x_2) \quad \therefore g: A \rightarrow B$ 是一个良定义的映射

又 $\forall x_1, x_2 \in A$, 若 $g(x_1) = g(x_2)$. 则 $f(x_1) = f(x_2)$

$\because x_1, x_2 \in A, A \subseteq X \quad \therefore x_1, x_2 \in X. \quad \because f(x_1) = f(x_2)$ 且

$f: X \rightarrow Y$ 是一个单射 $\therefore x_1 = x_2 \quad \therefore g: A \rightarrow B$ 是一个单射.

又 $\forall \beta \in B. \quad \therefore B = \{f(x) \mid x \in A\}. \quad \beta \in B.$

$\therefore \exists \alpha \in A, \text{ s.t. } \beta = f(\alpha). \quad \therefore \alpha \in A.$

$\therefore g(\alpha) = f(\alpha) = \beta. \quad \therefore g: A \rightarrow B$ 是一个满射.

$\therefore g: A \rightarrow B$ 是一个双射. $\square$

~~Lemma: 映射 π~~

Lemma: 又 $\forall n \in \mathbb{Z}_{\geq 1}$, 有: 映射 $\pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是双射.
$$x \mapsto [x]_n$$

Proof: 又 $\forall x \in \{1, 2, \dots, n\}$, 有: $\pi(x) = [x]_n \in \mathbb{Z}/n\mathbb{Z}$

又 $\forall x_1, x_2 \in \{1, 2, \dots, n\}$,

若 $x_1 = x_2$, 则 $x_1 \equiv x_2 \pmod{n} \quad \therefore [x_1]_n = [x_2]_n$

$\therefore \pi(x_1) = [x_1]_n = [x_2]_n = \pi(x_2)$

$\therefore \pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是一个良定义的映射.

若 $\pi(x_1) = \pi(x_2)$, 则 ~~$[x_1]_n = [x_2]_n$~~ $[x_1]_n = [x_2]_n$

$\therefore x_1 \equiv x_2 \pmod{n} \quad \therefore n \mid x_1 - x_2.$

$\because -(n-1) \leq x_1 - x_2 \leq n-1 \quad \therefore x_1 - x_2 = 0 \quad \therefore x_1 = x_2$

$\therefore \pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是一个单射.

对 $\forall [x]_n \in \mathbb{Z}/n\mathbb{Z}$ (其中 $x \in \mathbb{Z}$), $\because x \in \mathbb{Z}, n \in \mathbb{Z}_{\geq 1}$

$\therefore$ 存在唯一的 $q, r \in \mathbb{Z}$, s.t. $x = nq + r, 0 \leq r < n$

如果 $r=0$, 则 $x = nq. \therefore x - n = nq - n = n(q-1)$

$\therefore n \mid x - n \therefore [x]_n = [n]_n \therefore \pi(n) = [n]_n = [x]_n.$

如果 $r \neq 0$, 则 $r \in \{1, \dots, n-1\} \therefore r \in \{1, \dots, n\}.$

$\therefore x - r = nq \therefore n \mid x - r \therefore x \equiv r \pmod{n}$

$\therefore [x]_n = [r]_n \therefore \pi(r) = [r]_n = [x]_n$

$\therefore \pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是满射.

$\therefore \pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是一个双射. $\square$

Remark: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, $\{1, 2, \dots, n\}$ 和 $\mathbb{Z}/n\mathbb{Z}$ 是两个集合,

$\pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是双射.

$$x \mapsto [x]_n$$

定义映射 $\Pi: P(\{1, 2, \dots, n\}) \rightarrow P(\mathbb{Z}/n\mathbb{Z})$ 则已经证明

$$A \mapsto \{\pi(x) \mid x \in A\}.$$

映射 Π 良定义且是双射.

考虑 $C = \{S \in P(\{1, 2, \dots, n\}) \mid \sum_{x \in S} x \equiv 0 \pmod{n}\}.$

$$D = \{T \in P(\mathbb{Z}/n\mathbb{Z}) \mid \sum_{y \in T} y = [0]_n\}.$$

$$\therefore C \subseteq P(\{1, 2, \dots, n\}), D \subseteq P(\mathbb{Z}/n\mathbb{Z})$$

任取 $\Pi(C)$ 中的一元: $\Pi(S)$ (其中 $S \in C$)

$$\because S \in C \quad \therefore S \in P(\{1, 2, \dots, n\}) \quad \therefore S \subseteq \{1, 2, \dots, n\}$$

$$\therefore \Pi(S) = \{\pi(x) \mid x \in S\} \subseteq \mathbb{Z}/n\mathbb{Z} \quad \therefore \Pi(S) \in P(\mathbb{Z}/n\mathbb{Z})$$

$$\because S \in C \quad \therefore \sum_{x \in S} x \equiv 0 \pmod{n}$$

这个等式成立需要用到 $\pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是单射.

$$\therefore \sum_{y \in \Pi(S)} y \stackrel{\uparrow}{=} \sum_{x \in S} \pi(x) = \sum_{x \in S} [x]_n$$

$$= \left[\sum_{x \in S} x \right]_n = [0]_n \quad \therefore \Pi(S) \in D. \quad \therefore \Pi(C) \subseteq D.$$

任取 D 中的一元: T (其中 $T \in P(\mathbb{Z}/n\mathbb{Z})$ 且 $\sum_{y \in T} y = [0]_n$)

$$\therefore \Pi: P(\{1, 2, \dots, n\}) \rightarrow P(\mathbb{Z}/n\mathbb{Z}) \text{ 是双射}$$

$$\therefore \Pi: P(\{1, 2, \dots, n\}) \rightarrow P(\mathbb{Z}/n\mathbb{Z}) \text{ 是满射}$$

$$\because T \in P(\mathbb{Z}/n\mathbb{Z}) \quad \therefore \exists S \in P(\{1, 2, \dots, n\}), \text{ s.t. } \Pi(S) = T$$

$$\therefore S \in P(\{1, 2, \dots, n\}) \quad \therefore S \subseteq \{1, 2, \dots, n\}$$

$$\therefore \Pi(S) = \{\pi(x) \mid x \in S\}$$

$$\because T \in D \quad \therefore \sum_{y \in T} y = [0]_n$$

这个等式成立需要用到 $\pi: \{1, 2, \dots, n\} \rightarrow \mathbb{Z}/n\mathbb{Z}$ 是单射

$$\therefore [0]_n = \sum_{y \in T} y = \sum_{y \in \Pi(S)} y \stackrel{\uparrow}{=} \sum_{x \in S} \pi(x) = \sum_{x \in S} [x]_n$$

$$= \left[\sum_{x \in S} x \right]_n \quad \therefore \sum_{x \in S} x \equiv 0 \pmod{n}$$

$$\therefore S \in P(\{1, 2, \dots, n\}) \text{ 且 } \sum_{x \in S} x \equiv 0 \pmod{n} \quad \therefore S \in C$$

$$\therefore T = \Pi(S) \in \Pi(C) \quad \therefore T \in \Pi(C) \quad \therefore D \subseteq \Pi(C)$$

$$\therefore \pi(C) = D$$

$\therefore$ 定义映射 $g: C \rightarrow D$ 则映射 $g: C \rightarrow D$ 良定且是
 $S \mapsto \pi(S)$.

双射.

$$\therefore |C| = |D|$$

$$\therefore (\mathbb{Z}/n\mathbb{Z} \text{ 中元素和为 } 0_{\mathbb{Z}/n\mathbb{Z}} \text{ 的子集个数}) = |D| = |C|$$

$$= (\{1, 2, \dots, n\} \text{ 中元素和为 } n \text{ 的非负整数倍子的子集个数}) \quad \square$$

Lemma: $n \in \mathbb{Z}_{\geq 1}$, $f(X) = \prod_{k=1}^n (1+X^k) \in \mathbb{C}[X]$.

$\deg(f) = 1+2+\dots+n = \frac{n(n+1)}{2} \in \mathbb{Z}_{\geq 1}$. 则已经证明了存在
 $a_0, a_1, \dots, a_N \in \mathbb{C}$ (其中 $N = \frac{n(n+1)}{2} \in \mathbb{Z}_{\geq 1}$), s.t. $f(X) = \sum_{m=0}^N a_m X^m$

~~对~~ 对 $\forall k \in \mathbb{Z}_{\geq 0}$, 定义 $E_k = \{S \in P(\{1, 2, \dots, n\}) \mid \sum_{x \in S} x = k\}$.

则有: 对 $\forall k \in \mathbb{Z}_{\geq 0}$, $a_k = |E_k|$.

Proof: $\therefore n \in \mathbb{Z}_{\geq 1}$, $f(X) = \prod_{k=1}^n (1+X^k) \in \mathbb{C}[X]$.

$$\therefore f(X) = \prod_{k=1}^n (1+X^k) = \sum_{S \subseteq \{1, 2, \dots, n\}} X^{\sum_{x \in S} x}$$

对 $\forall k \in \mathbb{Z}_{\geq 0}$, $\therefore E_k = \{S \in P(\{1, 2, \dots, n\}) \mid \sum_{x \in S} x = k\}$ ~~$E_0 = \emptyset$~~

$\therefore E_k \subseteq P(\{1, 2, \dots, n\}) \quad \therefore P(\{1, 2, \dots, n\}) = \bigsqcup_{k \in \mathbb{Z}_{\geq 0}} E_k$

$$\therefore f(X) = \prod_{k=1}^n (1+X^k) = \sum_{S \subseteq \{1,2,\dots,n\}} X^{\sum_{x \in S} x} = \sum_{S \in P(\{1,2,\dots,n\})} X^{\sum_{x \in S} x}$$

$$= \sum_{k \in \mathbb{Z}_{\geq 0}} \left(\sum_{S \in E_k} X^{\sum_{x \in S} x} \right) = \sum_{k \in \mathbb{Z}_{\geq 0}} \left(\sum_{S \in E_k} X^k \right)$$

~~$$= \sum_{k \in \mathbb{Z}_{\geq 0}} \left(\sum_{S \in E_k} X^k \right) = \sum_{k \in \mathbb{Z}_{\geq 0}} |E_k| \cdot X^k = \sum_{m=0}^N a_m X^m.$$~~

(注意: $E_0 = \{\emptyset\}$, $E_N = E_{\frac{n(n+1)}{2}} = \{\{1,2,\dots,n\}\}$.)

$\therefore$ 对 $\forall k \in \mathbb{Z}_{\geq 0}$, 有: $a_k = |E_k|$. $\square$

Remark: $C = \{S \in P(\{1,2,\dots,n\}) \mid \sum_{x \in S} x \equiv 0 \pmod{n}\}$
 $= \{S \in P(\{1,2,\dots,n\}) \mid \exists j \in \mathbb{Z}_{\geq 0}, \text{ s.t. } \sum_{x \in S} x = jn\}$

$$\therefore C = \bigsqcup_{j \in \mathbb{Z}_{\geq 0}} E_{jn}.$$

(对 $\forall S \in$ 右边, 有: $\exists j \in \mathbb{Z}_{\geq 0}$, s.t. $S \in E_{jn}$, $jn \in \mathbb{Z}_{\geq 0}$)

$\therefore S \in P(\{1,2,\dots,n\})$ 且 $\sum_{x \in S} x = jn$. $\therefore S \in C$ $\therefore$ 右边 $\subseteq$ 左边.

对 $\forall S \in$ 左边, 有: $S \in P(\{1,2,\dots,n\})$ 且 $\exists j \in \mathbb{Z}_{\geq 0}$, s.t. $\sum_{x \in S} x = jn$.

$\therefore j \in \mathbb{Z}_{\geq 0}$, $n \in \mathbb{Z}_{\geq 1}$ $\therefore jn \in \mathbb{Z}_{\geq 0}$.

$\therefore S \in P(\{1,2,\dots,n\})$ 且 $\sum_{x \in S} x = jn \in \mathbb{Z}_{\geq 0}$ $\therefore S \in E_{jn}$. $\therefore S \in$ 右边

$\therefore$ 左边 $\subseteq$ 右边. $\therefore$ 左边 = 右边.

对 $\forall j_1, j_2 \in \mathbb{Z}_{\geq 0}$, $j_1 \neq j_2$. 假设 $E_{j_1 n} \cap E_{j_2 n} \neq \emptyset$. 则

$$\exists S \in E_{j_1 n} \cap E_{j_2 n} \quad \therefore S \in E_{j_1 n} \text{ 且 } S \in E_{j_2 n}.$$

$$\therefore S \in E_{j_1 n} \quad \therefore S \in P(\{1, 2, \dots, n\}) \text{ 且 } \sum_{x \in S} x = j_1 n \in \mathbb{Z}_{\geq 0}$$

$$\therefore S \in E_{j_2 n} \quad \therefore S \in P(\{1, 2, \dots, n\}) \text{ 且 } \sum_{x \in S} x = j_2 n \in \mathbb{Z}_{\geq 0}$$

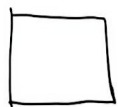
$$\therefore j_1 n = j_2 n \quad \therefore j_1 = j_2. \text{ 矛盾. } \therefore E_{j_1 n} \cap E_{j_2 n} = \emptyset.$$

$$\therefore C = \bigsqcup_{j \in \mathbb{Z}_{\geq 0}} E_{jn}.$$

$$\therefore |C| = \sum_{j \in \mathbb{Z}_{\geq 0}} |E_{jn}| = \sum_{j \in \mathbb{Z}_{\geq 0}} a_{jn}$$

$$\therefore (\mathbb{Z}/n\mathbb{Z} \text{ 中元素和为 } 0_{\mathbb{Z}/n\mathbb{Z}} \text{ 的子集个数}) = |C| = \sum_{j \in \mathbb{Z}_{\geq 0}} a_{jn}$$

$$= \frac{1}{n} \sum_{j=1}^n f(\zeta^j)$$



前方补充的笔记完.

后面是正式笔记.

$$\mathbb{C} = \{x + yi \mid x \in \mathbb{R} \text{ 且 } y \in \mathbb{R}\}.$$

定义 $\mathbb{C}$ 上的加法和乘法为: 对 $\forall z_1 = x_1 + y_1 i, z_2 = x_2 + y_2 i \in \mathbb{C}$ (其中 $x_1, y_1, x_2, y_2 \in \mathbb{R}$), 定义

$$z_1 + z_2 := (x_1 + x_2) + (y_1 + y_2)i \in \mathbb{C}$$

$$z_1 \cdot z_2 = (x_1 x_2 - y_1 y_2) + (x_1 y_2 + x_2 y_1)i \in \mathbb{C}$$

不难证明, $(\mathbb{C}, +, \cdot, 0+0i, 1+0i)$ 是域.

$\therefore (\mathbb{C}^\times, \cdot)$ 是一个交换群. $\mathbb{C}^\times = \mathbb{C} \setminus \{0+0i\}$

$$\text{对 } \forall n \in \mathbb{Z}_{\geq 1}, \mu_n(\mathbb{C}) = \{z \in \mathbb{C}^\times : z^n = 1\} = \{z \in \mathbb{C} : z^n = 1+0i\}$$

已经证明 $\mu_n(\mathbb{C})$ 是群 $(\mathbb{C}^\times, \cdot)$ 的子群. 集合 $\mu_n(\mathbb{C})$ 中的元素称为域 $\mathbb{C}$ 中的 n 次单位根.

$$\because \mathbb{C} \text{ 是域}, n \in \mathbb{Z}_{\geq 1} \quad \therefore |\mu_n(\mathbb{C})| \leq n$$

$$\because \mathbb{C} \text{ 是域}, \mu_n(\mathbb{C}) \text{ 是群 } (\mathbb{C}^\times, \cdot) \text{ 的子群}, |\mu_n(\mathbb{C})| \leq n$$

$$\therefore \mu_n(\mathbb{C}) \text{ 是群 } (\mathbb{C}^\times, \cdot) \text{ 的有限子群} \quad \therefore \mu_n(\mathbb{C}) \text{ 是循环群}$$

$$\therefore (\mu_n(\mathbb{C}), \cdot) \text{ 是循环群.}$$

总结一下上面的讨论: $(\mathbb{C}, +, \cdot, 0+0i, 1+0i)$ 是域, 对 $\forall n \in \mathbb{Z}_{\geq 1}$, $(\mu_n(\mathbb{C}), \cdot)$ 是交换群 $(\mathbb{C}^\times, \cdot)$ 的有限子群, $(\mu_n(\mathbb{C}), \cdot)$ 是循环群

$$\mu_n(\mathbb{C}) = \{z \in \mathbb{C}^\times : z^n = 1+0i\} = \{z \in \mathbb{C} : z^n = 1+0i\}$$

$$\text{对 } \forall n \in \mathbb{Z}_{\geq 1}, e^{\frac{2\pi}{n}i} = e^{i\frac{2\pi}{n}} = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n} \in \mathbb{C}^\times \text{ 且}$$

$$(e^{\frac{2\pi}{n}i})^n = (e^{i\frac{2\pi}{n}})^n = e^{i(n\frac{2\pi}{n})} = e^{i \cdot 2\pi} = \cos 2\pi + i \sin 2\pi = 1+0i = 1$$

$\therefore e^{\frac{2\pi}{n}i} \in \mu_n(\mathbb{C}) \quad \therefore (\mu_n(\mathbb{C}), \cdot)$ 是交换群, $e^{\frac{2\pi}{n}i} \in \mu_n(\mathbb{C})$.

$\therefore n \in \mathbb{Z}_{\geq 1}$ 且 $(e^{\frac{2\pi}{n}i})^n = 1 + 0i = 1 \quad \therefore \text{ord}(e^{\frac{2\pi}{n}i}) \in \mathbb{Z}_{\geq 1}$.

对 $\forall k \in \mathbb{Z}_{\geq 1}$, 有: $(e^{\frac{2\pi}{n}i})^k = 1 + 0i \Leftrightarrow e^{\frac{2k\pi}{n}i} = 1 + 0i$

$\Leftrightarrow \cos \frac{2k\pi}{n} + i \sin \frac{2k\pi}{n} = 1 + 0i \Leftrightarrow \cos \frac{2k\pi}{n} = 1$ 且 $\sin \frac{2k\pi}{n} = 0$

$\Leftrightarrow \frac{k}{n} \in \mathbb{Z} \Leftrightarrow n | k$

$\therefore \text{ord}(e^{\frac{2\pi}{n}i}) = \min \{ k \in \mathbb{Z}_{\geq 1} : (e^{\frac{2\pi}{n}i})^k = 1 + 0i \} = \min \{ k \in \mathbb{Z}_{\geq 1} : n | k \} = n$

$\therefore e^{\frac{2\pi}{n}i} \in \mu_n(\mathbb{C})$ 且 $n = \text{ord}(e^{\frac{2\pi}{n}i}) = |\langle e^{\frac{2\pi}{n}i} \rangle|$

$\therefore \langle e^{\frac{2\pi}{n}i} \rangle \leq \mu_n(\mathbb{C}) \quad \therefore n = \text{ord}(e^{\frac{2\pi}{n}i}) = |\langle e^{\frac{2\pi}{n}i} \rangle| \leq |\mu_n(\mathbb{C})| \leq n$

$\therefore |\mu_n(\mathbb{C})| = |\langle e^{\frac{2\pi}{n}i} \rangle| = n \quad \therefore (\mu_n(\mathbb{C}), \cdot)$ 是 n 阶循环群,

且 $\mu_n(\mathbb{C}) = \langle e^{\frac{2\pi}{n}i} \rangle$.

对 $\forall z \in \mathbb{C}$, 若 $z^n = 1$, 则有 $z \in \{ e^{\frac{2k\pi}{n}i} \mid k=0, 1, \dots, n-1 \}$

若 $z \in \{ e^{\frac{2k\pi}{n}i} \mid k=0, 1, \dots, n-1 \}$, 则 $z^n = 1$

$\therefore$ 对 $\forall z \in \mathbb{C}$, 有: $z^n = 1 \Leftrightarrow z \in \{ e^{\frac{2k\pi}{n}i} \mid k=0, 1, \dots, n-1 \}$.

$\therefore \mu_n(\mathbb{C}) = \{ z \in \mathbb{C} : z^n = 1 \} = \{ z \in \mathbb{C} : z = e^{\frac{2k\pi}{n}i}, k=0, 1, \dots, n-1 \}$

$= \{ e^{\frac{2k\pi}{n}i} \mid k=0, 1, \dots, n-1 \} = \{ (e^{\frac{2\pi}{n}i})^k \mid k=0, 1, \dots, n-1 \}$

$\therefore (\mu_n(\mathbb{C}), \cdot)$ 是交换群, $e^{\frac{2\pi}{n}i} \in \mu_n(\mathbb{C})$, $\text{ord}(e^{\frac{2\pi}{n}i}) = n \in \mathbb{Z}_{\geq 1}$

则对 $\forall k \in \mathbb{Z}$, 有: $\text{ord}((e^{\frac{2\pi}{n}i})^k) = \frac{n}{\gcd(k, n)}$

$\therefore$ 对 $\forall k \in \mathbb{Z}$, 有: $\text{ord}((e^{\frac{2\pi}{n}i})^k) = n \Leftrightarrow \frac{n}{\gcd(k, n)} = n \Leftrightarrow \gcd(k, n) = 1$

我已经证明了如下结论:

定理(复数域中的 n 次单位原根) $(\mathbb{C}, +, \cdot, 0+0i, 1+0i)$ 是域,
对 $\forall n \in \mathbb{Z}_{\geq 1}$, 已经证明了 $(\mathbb{C}^\times, \cdot)$ 是交换群, $(\mu_n(\mathbb{C}), \cdot)$ 是交换群 $(\mathbb{C}^\times, \cdot)$
的有限子群, $(\mu_n(\mathbb{C}), \cdot)$ 是循环群, $\mu_n(\mathbb{C}) = \{z \in \mathbb{C}^\times : z^n = 1+0i\} = \{z \in \mathbb{C} : z^n = 1+0i\}$
已经证明了 $(\mu_n(\mathbb{C}), \cdot)$ 是 n 阶循环群且 $\mu_n(\mathbb{C}) = \langle e^{\frac{2\pi i}{n}} \rangle$, 且有
 $\mu_n(\mathbb{C}) = \{e^{\frac{2k\pi i}{n}} \mid k=0, 1, \dots, n-1\} = \{(e^{\frac{2\pi i}{n}})^k \mid k=0, 1, \dots, n-1\}$. 则有:

对 $\forall k \in \mathbb{Z}$, $e^{\frac{2k\pi i}{n}}$ 是域 $\mathbb{C}$ 中的 n 次单位原根

$$\Leftrightarrow \text{ord}(e^{\frac{2k\pi i}{n}}) = n$$

$$\Leftrightarrow \gcd(k, n) = 1 \Leftrightarrow (\mathbb{Z}/n\mathbb{Z}, +) = \langle [k]_n \rangle$$

$$\Leftrightarrow \mu_n(\mathbb{C}) = \langle e^{\frac{2k\pi i}{n}} \rangle \simeq (\mathbb{Z}/n\mathbb{Z}, +)$$

$$\Leftrightarrow e^{\frac{2k\pi i}{n}} \text{ 是 } n\text{ 阶循环群 } (\mu_n(\mathbb{C}), \cdot) \text{ 的生成元.}$$

~~推论: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 复数域 $\mathbb{C}$ 中的 n 次单位原根 共有 $\varphi(n)$ 个.~~

~~Proof: 显然~~

~~Lemma: 对 $\forall z \in \mathbb{C}$, 有: $z^n = 1$~~

Lemma: 对 $\forall z \in \mathbb{C}$, 有: $z^0 = 1 \Leftrightarrow z \in \mathbb{C} \setminus \{0\} = \mathbb{C}^\times$

Proof: $(\Rightarrow) : \because z \in \mathbb{C}$ 且 $z^0 = 1 \therefore z^0$ 有定义 $\therefore z \neq 0 \therefore z \in \mathbb{C} \setminus \{0\}$

$(\Leftarrow) : \because z \in \mathbb{C} \setminus \{0\} \therefore z \neq 0 \therefore z^0 = 1$

Lemma: 对 $\forall z \in \mathbb{C}$, $\forall n \in \mathbb{Z} \setminus \{0\}$, 有:

$$z^n = 1 \Leftrightarrow z \in \{e^{\frac{2k\pi i}{n}} \mid k \in \mathbb{Z}\}$$

Proof: ($\Rightarrow$): 已知 $z \in \mathbb{C}$, $n \in \mathbb{Z} \setminus \{0\}$, $z^n = 1$

假设 $z=0$, 则若 $n > 0$, 则 $n \in \mathbb{Z}_{\geq 1}$ $\therefore 1 = z^n = 0^n = 0$ 矛盾.

若 $n < 0$, 则 $n \in \mathbb{Z}_{\leq -1}$ $\therefore -n \in \mathbb{Z}_{\geq 1}$

$$\therefore z^n = z^{-(-n)} = \frac{1}{z^{-n}} = \frac{1}{0^{-n}} = \frac{1}{0} \quad \therefore 1 = z^n = \frac{1}{0} \text{ 矛盾.}$$

$\therefore z \neq 0 \quad \therefore z \in \mathbb{C} \setminus \{0\}$. 设 $z = re^{i\theta}$ (其中 $r \in \mathbb{R}_{>0}$, $\theta \in \mathbb{R}$)

$$\therefore z^n = 1 \quad \therefore (re^{i\theta})^n = 1 \xrightarrow{\text{De Moivre's formula}} r^n e^{i(n\theta)} = 1 \quad (\text{注意: 对 } \forall n \in \mathbb{Z} \setminus \{0\})$$

$$\therefore \begin{cases} r^n = 1 \\ n\theta = 2k\pi, k \in \mathbb{Z} \end{cases} \quad \therefore \begin{cases} r = 1 \\ \theta = \frac{2k\pi}{n}, k \in \mathbb{Z} \end{cases}$$

$$\therefore z = e^{i \frac{2k\pi}{n}}, k \in \mathbb{Z} \quad \therefore z \in \left\{ e^{i \frac{2k\pi}{n}} \mid k \in \mathbb{Z} \right\}.$$

($\Leftarrow$): 若 $z \in \left\{ e^{i \frac{2k\pi}{n}} \mid k \in \mathbb{Z} \right\}$. 则 $\exists k \in \mathbb{Z}$, s.t. $z = e^{i \frac{2k\pi}{n}}$

$$\therefore z^n = \left(e^{i \frac{2k\pi}{n}} \right)^n = e^{i(n \cdot \frac{2k\pi}{n})} = e^{i(2k\pi)} = \cos(2k\pi) + i \sin(2k\pi) \\ = 1 + 0i = 1 \quad \square$$

$$\text{Lemma: 对 } \forall n \in \mathbb{Z}_{\geq 1}, \text{ 有: } \mu_n(\mathbb{C}) = \left\{ e^{i \frac{2k\pi}{n}} \mid k \in \mathbb{Z} \right\} = \left\{ e^{i \frac{2k\pi}{n}} \mid k=0, 1, \dots, n-1 \right\} \\ = \left\{ e^{i \frac{2k\pi}{n}} \mid k=1, 2, \dots, n \right\}$$

Proof: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 有: $\mu_n(\mathbb{C}) = \{z \in \mathbb{C}^\times : z^n = 1\} = \{z \in \mathbb{C} : z^n = 1\}$

$$= \{z \in \mathbb{C} : z = e^{i \frac{2k\pi}{n}}, k \in \mathbb{Z}\} = \left\{ e^{i \frac{2k\pi}{n}} \mid k \in \mathbb{Z} \right\}.$$

$$\text{对 } \forall k_1, k_2 \in \mathbb{Z}, \text{ 有: } e^{i \frac{2k_1\pi}{n}} = e^{i \frac{2k_2\pi}{n}} \Leftrightarrow \frac{2k_1\pi}{n} - \frac{2k_2\pi}{n} \in 2\pi\mathbb{Z}$$

$$\Leftrightarrow n \mid k_1 - k_2 \Leftrightarrow k_1 \equiv k_2 \pmod{n}$$

$$\begin{aligned}\therefore \mu_n(\mathbb{C}) &= \left\{ e^{\frac{2k\pi}{n}i} \mid k \in \mathbb{Z} \right\} = \left\{ e^{\frac{2k\pi}{n}i} \mid k=0,1,\dots,n-1 \right\} \\ &= \left\{ e^{\frac{2k\pi}{n}i} \mid k=1,2,\dots,n \right\}\end{aligned}$$

Lemma: 对 $\forall n \in \mathbb{Z}_{\geq 1}$, 复数域 $\mathbb{C}$ 中的 n 次单位原根一共有 $\varphi(n)$ 个.

Proof: 对 $\forall k \in \mathbb{Z}$, $e^{\frac{2k\pi}{n}i}$ 是复数域 $\mathbb{C}$ 中的 n 次单位原根

$$\Leftrightarrow \gcd(k, n) = 1$$

$\therefore$ 对 $\forall k \in \{1, 2, \dots, n\}$, $e^{\frac{2k\pi}{n}i}$ 是复数域 $\mathbb{C}$ 中的 n 次单位原根

$$\Leftrightarrow \gcd(k, n) = 1$$

$\Leftrightarrow k$ 是不超过 n 且与 n 互素的正整数

$\therefore$ 复数域 $\mathbb{C}$ 中的 n 次单位原根有 $\varphi(n)$ 个 $\square$

Lemma: $n \in \mathbb{Z}_{\geq 1}$, $\zeta \in \mathbb{C}^\times$ 为任意的一个 n 次单位原根. 则对 $\forall k \in \mathbb{Z}$,

$$\text{有: } \sum_{j=1}^n \zeta^{kj} = \begin{cases} n, & \text{当 } n \mid k \text{ 时} \\ 0, & \text{当 } n \nmid k \text{ 时} \end{cases}$$

Proof: $\because \zeta \in \mathbb{C}^\times$ 是一个 n 次单位原根 $\therefore \exists \lambda \in \mathbb{Z}$, s.t. $\zeta = e^{\frac{2\lambda\pi}{n}i}$,

$$\text{且 } \gcd(\lambda, n) = 1$$

$\therefore$ 对 $\forall k \in \mathbb{Z}$, $\forall j \in \{1, 2, \dots, n\}$, 有: $kj \in \mathbb{Z}$

$$\therefore \zeta^{kj} = \left(e^{\frac{2\lambda\pi}{n}i} \right)^{kj} = e^{\frac{2\lambda kj\pi}{n}i}$$

$$\text{当 } n \mid k \text{ 时, } \frac{k}{n} \in \mathbb{Z} \quad \therefore \frac{2\lambda kj\pi}{n} = 2 \cdot (\lambda \cdot \frac{k}{n} \cdot j) \pi \in 2\pi\mathbb{Z}$$

$$\therefore \zeta^{kj} = e^{\frac{2\lambda kj\pi}{n}i} = 1 \quad \therefore \sum_{j=1}^n \zeta^{kj} = \sum_{j=1}^n 1 = n.$$

当 $n \nmid k$ 时, 假设 $\zeta^k = 1$, 则有: $e^{\frac{2\lambda k \pi}{n} i} = 1$

$$\therefore \frac{2\lambda k \pi}{n} \in 2\pi \mathbb{Z} \quad \therefore \lambda \cdot \frac{k}{n} \in \mathbb{Z} \quad \therefore n \mid \lambda k.$$

$$\therefore n \in \mathbb{Z}_{\geq 1}, \lambda \in \mathbb{Z}, k \in \mathbb{Z}, \gcd(\lambda, n) = \gcd(n, \lambda) = 1 \quad \therefore n \mid k.$$

矛盾. $\zeta^k \neq 1$.

$$\therefore \sum_{j=1}^n \zeta^{kj} = \sum_{j=1}^n (\zeta^k)^j = \frac{\zeta^k - (\zeta^k)^{n+1}}{1 - \zeta^k} = \frac{\zeta^k - (\zeta^k)^n \cdot \zeta^k}{1 - \zeta^k}$$

$$\therefore (\zeta^k)^n = \zeta^{kn} = \left(e^{\frac{2\lambda \pi}{n} i} \right)^{kn} = e^{(2\lambda k \pi) i} = 1$$

$$\therefore \sum_{j=1}^n \zeta^{kj} = 0 \quad \square$$

Lemma: $\mathbb{C}$ 是复数域. 已经证明 $\mathbb{C}[X]$ 是 Euclid 环, 主理想环, 唯一分解环. 定义 $f = \prod_{k=1}^n (1+X^k) \in \mathbb{C}[X]$ ^($n \in \mathbb{Z}_{\geq 1}$). 设 f 展开后为 $f = \sum_{k \geq 0} a_k X^k$

(对 $\forall k \in \mathbb{Z}_{\geq 0}, a_k \in \mathbb{C}$). 则对复数域 $\mathbb{C}$ 中的任意 n 次单位原根 $\zeta \in \mathbb{C}^\times$, 有:

$$\frac{1}{n} \sum_{j=1}^n f(\zeta^j) = \sum_{j \geq 0} a_j$$

$$\begin{aligned} \text{Proof: 左边} &= \frac{1}{n} \sum_{j=1}^n f(\zeta^j) = \frac{1}{n} \sum_{j=1}^n \left(\sum_{k \geq 0} a_k (\zeta^j)^k \right) = \frac{1}{n} \sum_{j=1}^n \sum_{k \geq 0} a_k \zeta^{kj} \\ &= \frac{1}{n} \sum_{k \geq 0} \sum_{j=1}^n a_k \zeta^{kj} = \frac{1}{n} \sum_{k \geq 0} a_k \left(\sum_{j=1}^n \zeta^{kj} \right) \end{aligned}$$

↓
两个有限求和一定可交换, 当 $k > \frac{n(n+1)}{2}$ 时 $a_k = 0 \in \mathbb{C}$.

$$\therefore \text{当 } n \mid k \text{ 时, } \sum_{j=1}^n \zeta^{kj} = n. \quad \therefore k = 0, n, 2n, 3n, 4n, \dots, mn$$

$$\left(\text{其中 } m \in \mathbb{Z}_{\geq 0} \text{ 且 } mn \leq \frac{n(n+1)}{2} \right) \quad \therefore mn \leq \frac{n(n+1)}{2} \text{ 且 } n \in \mathbb{Z}_{\geq 1} \quad \therefore m \leq \frac{n+1}{2}$$

当 $n \in \mathbb{Z}_{\geq 1}$ 且 n 为奇数时, $\frac{n+1}{2} \in \mathbb{Z}_{\geq 1}$ $\therefore m = \frac{n+1}{2} = \lfloor \frac{n+1}{2} \rfloor$.

当 $n \in \mathbb{Z}_{\geq 1}$ 且 n 为偶数时, $\frac{n+1}{2} \notin \mathbb{Z}_{\geq 1}$. $\frac{n+1}{2} > 1$. $m = \lfloor \frac{n+1}{2} \rfloor$

$$\therefore \text{左边} = \frac{1}{n} \sum_{k \geq 0} a_k \left(\sum_{j=1}^n \zeta^{kj} \right) = \frac{1}{n} \sum_{t=0}^{\lfloor \frac{n+1}{2} \rfloor} a_{tn} \cdot n = \sum_{t=0}^{\lfloor \frac{n+1}{2} \rfloor} a_{tn} = \sum_{j=0}^{\lfloor \frac{n+1}{2} \rfloor} a_j = \text{右边}.$$

□

Lemma: $n \in \mathbb{Z}_{\geq 1}$, $\zeta \in \mathbb{C}^\times$ 是任意的一个 n 次单位原根. 对 $\forall j \in \mathbb{Z}_{\geq 1}$,

令 $d = \frac{n}{\gcd(j, n)}$. 则 ζ^j 是 d 次单位原根.

Proof: $\because j \in \mathbb{Z}_{\geq 1}$, $n \in \mathbb{Z}_{\geq 1}$ $\therefore \gcd(j, n) \in \mathbb{Z}_{\geq 1}$. 且 $\gcd(j, n) \mid n$

$$\therefore d = \frac{n}{\gcd(j, n)} \in \mathbb{Z}_{\geq 1}.$$

$\mathbb{C}$ 是域, $d \in \mathbb{Z}_{\geq 1}$, ~~已经证明~~ 已经证明了 $\mu_d(\mathbb{C})$ 是循环群,

$$\mu_d(\mathbb{C}) = \{r \in \mathbb{C}^\times : r^d = 1\}. \quad \because \zeta \in \mathbb{C}^\times \text{ 是 } n \text{ 次单位原根.}$$

$$\therefore \zeta \in \mathbb{C}^\times \text{ 且 } \zeta^n = 1$$

$\because \mathbb{C}$ 是域 $\therefore (\mathbb{C}^\times, \cdot)$ 是交换群. $\zeta \in \mathbb{C}^\times$. $\therefore \zeta \in \mathbb{C}^\times$ 是 n 次单位原根

$\therefore \text{ord}(\zeta) = n \in \mathbb{Z}_{\geq 1}$ 对于 $j \in \mathbb{Z}_{\geq 1}$, 有:

$$\text{ord}(\zeta^j) = \frac{n}{\gcd(j, n)} = d \in \mathbb{Z}_{\geq 1}$$

$$\because (\mathbb{C}^\times, \cdot) \text{ 是交换群, } \zeta^j \in \mathbb{C}^\times, \text{ord}(\zeta^j) = d \in \mathbb{Z}_{\geq 1} \quad \therefore (\zeta^j)^d = 1$$

$\therefore \zeta^j \in \mu_d(\mathbb{C})$. $\because \mathbb{C}$ 是域, $d \in \mathbb{Z}_{\geq 1}$ $\therefore \mu_d(\mathbb{C})$ 是交换群 $(\mathbb{C}^\times, \cdot)$ 的子群

$\therefore \zeta^j \in \mu_d(\mathbb{C})$ 且 $\text{ord}(\zeta^j) = d \in \mathbb{Z}_{\geq 1}$ $\therefore \zeta^j$ 是复数域 $\mathbb{C}$ 中的 d 次单位原根.

□

Lemma: $n \in \mathbb{Z}_{\geq 1}$, $\zeta \in \mathbb{C}^\times$ 是任意的一个 n 次单位原根. 对 $\forall j \in \mathbb{Z}_{\geq 1}$, 令 $d = \frac{n}{\gcd(j, n)}$. 则已经证明了 $d \in \mathbb{Z}_{\geq 1}$ 且 ζ^j 是复数域 $\mathbb{C}$ 中的 d 次单位原根. 则有: $\zeta^j, \zeta^{2j}, \dots, \zeta^{dj}$ 两两不同, 且都是多项式 $X^d - 1 \in \mathbb{C}[X]$ 的根.

Proof: 对 $\forall t \in \{1, 2, \dots, d\}$, 有: $\because \zeta^j$ 是复数域 $\mathbb{C}$ 中的 d 次单位原根

$$\therefore \zeta^j \in \mathbb{C}^\times \quad \therefore (\zeta^j)^t \in \mathbb{C}^\times \quad \therefore \zeta^{tj} \in \mathbb{C}^\times$$

$$\therefore (\zeta^{tj})^d = \zeta^{d tj} = (\zeta^{dj})^t = ((\zeta^j)^d)^t = 1^t = 1$$

$\therefore \zeta^{tj} \in \mathbb{C}^\times$ 是多项式 $X^d - 1 \in \mathbb{C}[X]$ 的根.

$\therefore \zeta^j, \zeta^{2j}, \dots, \zeta^{dj} \in \mathbb{C}^\times$ 是多项式 $X^d - 1 \in \mathbb{C}[X]$ 的根.

假设存在 $s, t \in \{1, 2, \dots, d\}$ 且 $s < t$, s.t. $\zeta^s = \zeta^t$

则有: ~~$\zeta^s = \zeta^t$~~ $\therefore \zeta^s = \zeta^t \in \mathbb{C}^\times \quad \therefore \zeta^{(t-s)} = \zeta^{t-s} = \frac{\zeta^t}{\zeta^s} = 1$

$$\therefore t-s \in \mathbb{Z}_{\geq 1} \text{ 且 } t-s < d \quad \therefore (\zeta^j)^{t-s} = 1$$

$$\therefore \text{ord}(\zeta^j) = d \in \mathbb{Z}_{\geq 1}$$

$$\therefore (\mathbb{C}^\times, \cdot) \text{ 是交换群, } \zeta^j \in \mathbb{C}^\times, \quad d = \text{ord}(\zeta^j) = \min \{k \in \mathbb{Z}_{\geq 1} : (\zeta^j)^k = 1\}$$

$$\therefore t-s \in \mathbb{Z}_{\geq 1}, (\zeta^j)^{t-s} = 1, \quad \therefore d \leq t-s < d \text{ 矛盾.}$$

$$\therefore \text{不存在 } s, t \in \{1, 2, \dots, d\} \text{ 且 } s < t, \text{ s.t. } \zeta^s = \zeta^t$$

$\therefore \zeta^j, \zeta^{2j}, \dots, \zeta^{dj} \in \mathbb{C}^\times$ 是多项式 $X^d - 1 \in \mathbb{C}[X]$ 的 d 个两两不同的根

□

Lemma: $n \in \mathbb{Z}_{\geq 1}$, $\zeta \in \mathbb{C}^\times$ 是任意的一个 n 次单位原根. 对 $\forall j \in \mathbb{Z}_{\geq 1}$, 令 $d = \frac{n}{\gcd(j, n)}$ 则已经证明了 $d \in \mathbb{Z}_{\geq 1}$ 且 ζ^j 是复数域 $\mathbb{C}$ 中的 d 次

单位原根. 已经证明了 $\zeta^j, \zeta^{2j}, \dots, \zeta^{dj} \in \mathbb{C}^\times$ 两两不同, 且都是多项式

$$X^d - 1 \in \mathbb{C}[X] \text{ 的根. 则有: } X^d - 1 = \prod_{k=1}^d (X - \zeta^{kj})$$

Proof: 考虑多项式 $H(X) = (X^d - 1) - \prod_{k=1}^d (X - \zeta^{kj}) \in \mathbb{C}[X]$.

假设 $H(X)$ 不是零多项式. 则 $H(X)$ 是非零多项式.

$$\deg(H(X)) \leq d-1$$

$\therefore \zeta^j, \zeta^{2j}, \dots, \zeta^{dj} \in \mathbb{C}^\times$ 两两不同, 且都是多项式 $H(X) \in \mathbb{C}[X]$ 的根.

$\therefore \mathbb{C}$ 是域, $H(X) \in \mathbb{C}[X]$ 是 $\mathbb{C}[X]$ 中的一个至多 $d-1$ 次的非零多项式.

$\therefore H(X)$ 至多只有 $d-1$ 个相异的根 (其中 $d-1 \in \mathbb{Z}_{\geq 0}$) 矛盾.

$$\therefore H(X) \text{ 是零多项式. } \therefore X^d - 1 = \prod_{k=1}^d (X - \zeta^{kj}) \quad \square$$

Lemma: $n \in \mathbb{Z}_{\geq 1}$, $\zeta \in \mathbb{C}^\times$ 是任意的一个 n 次单位原根. 对 $\forall j \in \mathbb{Z}_{\geq 1}$,

令 $d = \frac{n}{\gcd(j, n)}$ 则已经证明了 $d \in \mathbb{Z}_{\geq 1}$ 且 ζ^j 是复数域 $\mathbb{C}$ 中的 d 次

单位原根. 则有: $\prod_{k=1}^d (1 + \zeta^{kj}) = \begin{cases} 2, & d \text{ 为奇数} \\ 0, & d \text{ 为偶数} \end{cases}$

Proof: $\therefore X^d - 1 = \prod_{k=1}^d (X - \zeta^{kj}) \quad \therefore (-1)^d - 1 = \prod_{k=1}^d (-1 - \zeta^{kj})$

~~$\therefore (-1)^d - 1 = \prod_{k=1}^d (-1 - \zeta^{kj})$~~

$$\therefore (-1)^d - 1 = \prod_{k=1}^d (-1)(1 + \zeta^{kj}) = (-1)^d \prod_{k=1}^d (1 + \zeta^{kj})$$

$$\text{当 } d \text{ 为奇数时, } -2 = - \prod_{k=1}^d (1 + \zeta^{kj}) \quad \therefore \prod_{k=1}^d (1 + \zeta^{kj}) = 2$$

$$\text{当 } d \text{ 为偶数时, } 0 = 1 \cdot \prod_{k=1}^d (1 + \zeta^{kj}) \quad \therefore \prod_{k=1}^d (1 + \zeta^{kj}) = 0 \quad \square$$

Lemma: $n \in \mathbb{Z}_{\geq 1}$, $\zeta \in \mathbb{C}^*$ 是任意的一个 n 次单位原根. 对 $\forall j \in \mathbb{Z}_{\geq 1}$, 令 $d = \frac{n}{\gcd(j, n)}$. 则已经证明了 $d \in \mathbb{Z}_{\geq 1}$ 且 ζ^j 是复数域 $\mathbb{C}$ 中的 d 次单位原根. $f = \prod_{k=1}^n (1 + X^k) \in \mathbb{C}[X]$. 则有: $f(\zeta^j) = \begin{cases} 2^{\frac{n}{d}}, & d \text{ 是奇数} \\ 0, & d \text{ 是偶数} \end{cases}$

$$\text{Proof: } \because j \in \mathbb{Z}_{\geq 1} \text{ 且 } n \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd(j, n) \in \mathbb{Z}_{\geq 1} \text{ 且 } \gcd(j, n) \mid n$$

$$\therefore d = \frac{n}{\gcd(j, n)} \in \mathbb{Z}_{\geq 1} \quad \therefore \frac{n}{d} = \gcd(j, n) \in \mathbb{Z}_{\geq 1} \quad \therefore d \leq n \text{ 且 } d \mid n$$

$$\therefore f(\zeta^j) = \prod_{k=1}^n (1 + (\zeta^j)^k) = \prod_{a=0}^{\frac{n}{d}-1} \prod_{r=1}^d (1 + (\zeta^j)^{ad+r}) \quad (\text{其中 } \frac{n}{d} - 1 \in \mathbb{Z}_{\geq 0})$$

$$\begin{aligned} \text{①: 右边} &= \prod_{r=1}^d (1 + (\zeta^j)^r) \prod_{r=1}^d (1 + (\zeta^j)^{d+r}) \prod_{r=1}^d (1 + (\zeta^j)^{2d+r}) \cdots \prod_{r=1}^d (1 + (\zeta^j)^{(n-d)+r}) \\ &= \prod_{k=1}^n (1 + (\zeta^j)^k) = \text{左边} \end{aligned}$$

$$\begin{aligned} \because a \in \mathbb{Z}_{\geq 0} \quad \therefore (\zeta^j)^{ad+r} &= (\zeta^j)^{ad} \cdot (\zeta^j)^r = ((\zeta^j)^d)^a \cdot (\zeta^j)^r \\ &= 1^a \cdot (\zeta^j)^r = (\zeta^j)^r \end{aligned}$$

$$\therefore f(\zeta^j) = \prod_{a=0}^{\frac{n}{d}-1} \prod_{r=1}^d (1 + (\zeta^j)^r) = \left(\prod_{r=1}^d (1 + (\zeta^j)^r) \right)^{\frac{n}{d}} = \left(\prod_{r=1}^d (1 + \zeta^{rj}) \right)^{\frac{n}{d}}$$

$$= \left(\prod_{k=1}^d (1 + \zeta^{kj}) \right)^{\frac{n}{d}} = \begin{cases} 2^{\frac{n}{d}}, & d \text{ 为奇数} \\ 0, & d \text{ 为偶数} \end{cases} \quad \square$$

定理: ~~$\mathbb{Z}/n\mathbb{Z}$~~ $n \in \mathbb{Z}_{\geq 1}$, 则有:

$$(\mathbb{Z}/n\mathbb{Z} \text{ 中元素和为 } 0_{\mathbb{Z}/n\mathbb{Z}} \text{ 的子集个数}) = \frac{1}{n} \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1} \\ d \text{ 是奇数}}} \varphi(d) 2^{\frac{n}{d}}$$

(子集容许为空集. 空集的元素和定义为 $0_{\mathbb{Z}/n\mathbb{Z}}$)

Proof: 定义 ~~$f(X)$~~ $f(X) = \prod_{k=1}^n (1 + X^k) \in \mathbb{C}[X]$.

$$\therefore \deg(f) = 1 + 2 + \dots + n = \frac{n(n+1)}{2} \in \mathbb{Z}_{\geq 1}$$

$$\therefore \exists a_0, a_1, \dots, a_N \in \mathbb{C}, \text{ s.t. } f(X) = \sum_{m=0}^N a_m X^m. \text{ 其中 } N = \frac{n(n+1)}{2} \in \mathbb{Z}_{\geq 1}$$

$$\therefore f(X) = \prod_{k=1}^n (1 + X^k) = \sum_{S \subseteq \{1, \dots, n\}} X^{\sum_{k \in S} k}$$

$\therefore f(X)$ 展开式中 X^m 的系数就是满足 $\sum_{k \in S} k = m$ 的 $\{1, \dots, n\}$ 的子集 S 的个数. 当 $S = \emptyset$ 时, $X^{\sum_{k \in S} k} = X^0 = 1$

考虑映射 $\pi: \{1, 2, \dots, n\} \longrightarrow \mathbb{Z}/n\mathbb{Z}$. 显然 π 是双射.

$$x \longmapsto [x]_n$$

$$\text{又 } \forall S \subseteq \{1, 2, \dots, n\}, \text{ 有: } \sum_{x \in \pi(S)} x = \sum_{k \in S} [k]_n = \left[\sum_{k \in S} k \right]_n$$

$$\therefore \sum_{x \in \mathcal{P}(S)} x = [0]_n \Leftrightarrow \left[\sum_{k \in S} k \right]_n = \boxed{0} [0]_n \Leftrightarrow n \mid \sum_{k \in S} k$$

$$\Leftrightarrow \exists t \in \mathbb{Z}_{\geq 0}, \text{ s.t. } \sum_{k \in S} k = tn \quad (t=0 \text{ 对应于 } S=\emptyset)$$

$\therefore f(X)$ 展开式中 X^{tn} ($t \in \mathbb{Z}_{\geq 0}$) 的系数就是满足 $\sum_{k \in S} k = tn$ ($t \in \mathbb{Z}_{\geq 0}$)

的 $\{1, \dots, n\}$ 的子集 S 的个数 ($t=0$ 对应于 $S=\emptyset$)

~~就是满足元素和为 $[0]_n$ 的 $\mathbb{Z}/n\mathbb{Z}$ 的子集个数~~

$$\begin{aligned} \therefore (\mathbb{Z}/n\mathbb{Z} \text{ 中元素和为 } 0_{\mathbb{Z}/n\mathbb{Z}} \text{ 的子集个数}) &= \sum_{t \in \mathbb{Z}_{\geq 0}} a_{tn} = \sum_{j \in \mathbb{Z}_{\geq 0}} a_{jn} \\ &= \sum_{j=0}^{\lfloor \frac{n}{2} \rfloor} a_{jn} = \frac{1}{n} \sum_{j=1}^n f(G^j) \end{aligned}$$

~~$$\text{令 } C_d = \left\{ r \in \{1, 2, \dots, n\} : \frac{n}{\gcd(r, n)} = d \right\}$$~~

① 对 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d \mid n$, 定义 $J_d = \left\{ r \in \{1, 2, \dots, n\} : \frac{n}{\gcd(r, n)} = d \right\}$

$$\text{令 } C_d = |J_d|. \quad \text{则有: } \{1, 2, \dots, n\} = \bigsqcup_{\substack{d \mid n \\ d \in \mathbb{Z}_{\geq 1}}} J_d$$

对 $\forall r \in \{1, 2, \dots, n\}$. $\therefore r \in \mathbb{Z}_{\geq 1}$ 且 $n \in \mathbb{Z}_{\geq 1} \therefore \gcd(r, n) \in \mathbb{Z}_{\geq 1}$

$\therefore \gcd(r, n) \mid n \therefore \frac{n}{\gcd(r, n)} \in \mathbb{Z}_{\geq 1}$. 令 $d = \frac{n}{\gcd(r, n)} \in \mathbb{Z}_{\geq 1}$

$\therefore \frac{n}{d} = \gcd(r, n) \in \mathbb{Z}_{\geq 1} \therefore d \mid n \therefore d \mid n$ 且 $d \in \mathbb{Z}_{\geq 1}$

$\therefore r \in \{1, 2, \dots, n\}$ 且 $\frac{n}{\gcd(r, n)} = d$, $d \mid n$ 且 $d \in \mathbb{Z}_{\geq 1}$

$\therefore r \in J_d$, $d \mid n$ 且 $d \in \mathbb{Z}_{\geq 1} \therefore \{1, 2, \dots, n\} \subseteq \bigsqcup_{\substack{d \mid n \\ d \in \mathbb{Z}_{\geq 1}}} J_d$.

~~又 $\forall d | n, d \in \mathbb{Z}_{\geq 1}$~~ 又 $\forall r \in \text{右边}$, 有: $\exists d | n$ 且 $d \in \mathbb{Z}_{\geq 1}$, s.t.

$$r \in J_d \quad \therefore r \in \{1, 2, \dots, n\} \text{ 且 } \frac{n}{\gcd(r, n)} = d \quad \therefore r \in \{1, 2, \dots, n\}$$

$$\therefore \text{右边} \subseteq \{1, 2, \dots, n\} \quad \therefore \{1, 2, \dots, n\} = \bigsqcup_{\substack{d | n \\ d \in \mathbb{Z}_{\geq 1}}} J_d.$$

又 $\forall d_1, d_2 \in \mathbb{Z}_{\geq 1}$, ~~d_1, d_2~~ $d_1 | n, d_2 | n$, ~~$d_1 \neq d_2$~~

假设 $J_{d_1} \cap J_{d_2} \neq \emptyset$. 则 $\exists r \in J_{d_1}$ 且 $r \in J_{d_2}$

$$\therefore r \in \{1, \dots, n\} \text{ 且 } \frac{n}{\gcd(r, n)} = d_1 \quad r \in \{1, \dots, n\} \text{ 且 } \frac{n}{\gcd(r, n)} = d_2.$$

$$\therefore d_1 = d_2. \text{ 矛盾. } \therefore J_{d_1} \cap J_{d_2} = \emptyset.$$

$$\therefore \{1, 2, \dots, n\} = \bigsqcup_{\substack{d | n \\ d \in \mathbb{Z}_{\geq 1}}} J_d.$$

又 $\forall d \in \mathbb{Z}_{\geq 1}$ 且 $d | n$, 有: $\because d | n \quad \therefore \frac{n}{d} \in \mathbb{Z}_{\geq 1}$.

$$\because d \in \mathbb{Z}_{\geq 1} \quad \therefore 0 < \frac{1}{d} \leq 1 \quad \therefore \frac{n}{d} \leq n \quad \therefore \frac{n}{d} \in \{1, \dots, n\}.$$

$$\because \frac{n}{d} | n, \frac{n}{d} \in \mathbb{Z}_{\geq 1}, n \in \mathbb{Z}_{\geq 1} \quad \therefore \gcd\left(\frac{n}{d}, n\right) = \frac{n}{d}.$$

$$\therefore \frac{n}{\gcd\left(\frac{n}{d}, n\right)} = \frac{n}{\frac{n}{d}} = n \cdot \frac{d}{n} = d$$

$$\therefore \frac{n}{d} \in \{1, \dots, n\} \text{ 且 } \frac{n}{\gcd\left(\frac{n}{d}, n\right)} = d \quad \therefore \frac{n}{d} \in J_d \quad \therefore J_d \neq \emptyset.$$

$$\therefore \text{又 } \forall d \in \mathbb{Z}_{\geq 1} \text{ 且 } d | n, \text{ 有: } J_d \neq \emptyset$$

$$\therefore (\mathbb{Z}/n\mathbb{Z} \text{ 中元素和为 } 0_{\mathbb{Z}/n\mathbb{Z}} \text{ 的子集个数}) = \frac{1}{n} \sum_{j=1}^n f(\zeta^j) = \frac{1}{n} \sum_{\substack{d | n \\ d \in \mathbb{Z}_{\geq 1}}} \left(\sum_{r \in J_d} f(\zeta^r) \right)$$

$$= \frac{1}{n} \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1} \\ d \text{ 是奇数}}} |J_d| \cdot 2^{\frac{n}{d}} = \frac{1}{n} \sum_{\substack{d|n \\ d \in \mathbb{Z}_{\geq 1} \\ d \text{ 是奇数}}} \varphi(d) \cdot 2^{\frac{n}{d}}$$

